



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CCS339 Cryptocurrency and Blockchain Technologies

Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit III / Bitcoin Miner

Course Outcome: CO 3

Unit Learning Outcome: UO 3b

Activity Chosen: Picture Prompt

Justification:

- In this activity, students are shown an image without any explanation and are asked to identify and explain it, justifying their answers based on their existing knowledge. They might also be asked to write about the image using specific terms from a recent lecture or to name the processes and concepts depicted in the image. Bitcoin mining is a crucial aspect of blockchain technology, where miners validate transactions and secure the network by solving complex mathematical problems. By showing students an image related to Bitcoin mining (such as a mining rig, a blockchain diagram, or a data center), without explanation, and asking them to identify and explain it, the activity leverages visual learning to deepen their understanding.
- **Time Allotted for the Activity: 20 minutes**

Details of the Implementation:

- The instructor selected an image related to Bitcoin mining, such as a mining rig, blockchain diagram, or data center, and ensured it was clear and relevant to the lecture content.
- At the beginning of the session, the instructor briefly explained the objective of the activity, emphasizing its purpose to apply theoretical knowledge in a practical context without providing any background information about the image.
- The instructor displayed the selected image to the students using a local IP.
- Students were then divided into small groups of 2-3 members.
- Each group discussed their individual interpretations and jot down their thoughts, observations regarding what the image represented and how it related to Bitcoin mining.
- Each group presented their interpretation and explanation of the image to the class.
- After all groups had presented and discussed their interpretations, the instructor revealed the correct explanation of the image.

CO - PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO9	PSO1
CO 3	2	1	1	1	2	2	1

(1 - Low 2 - Moderate 3 - High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO5
	2	1	1	1	2
Justification for correlation	Students applied their understanding of blockchain technology, cryptographic principles, and the mathematics underlying Bitcoin mining algorithms to interpret the image.	Students analyzed the components and processes depicted in the Bitcoin miner image, identifying key elements and their functions in the context of blockchain technology.	Through group discussions, students conceptualized the design and operational aspects of Bitcoin miners, considering factors like energy consumption, efficiency, and security.	The activity encouraged students to investigate and interpret the technical and operational details of Bitcoin mining from the image.	Students discussed the use of modern tools and technologies like ASIC miners and blockchain software, understanding their roles and limitations in Bitcoin mining.
	PO9	PSO1			
	2	1			
	Students worked in groups to analyze and interpret the image, enhancing their teamwork and collaboration skills.	Students will continuously engage in lifelong learning and stay updated with the latest advancements in blockchain mining technique.			

- **Images / Screenshot of the practice:**

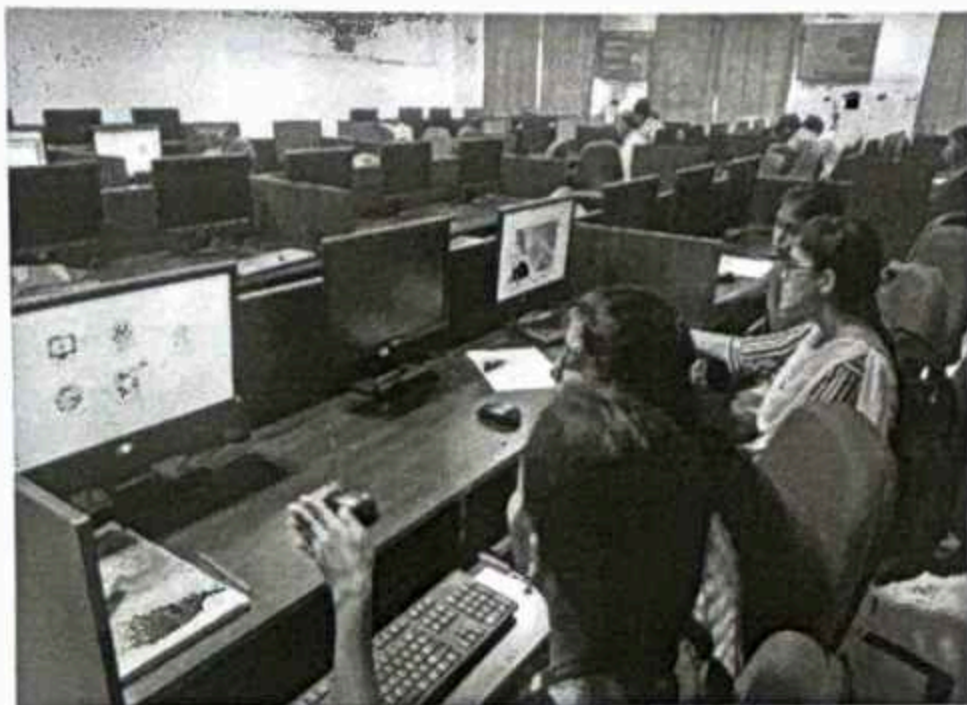


Fig:1 Glimpse of Picture Prompt activity

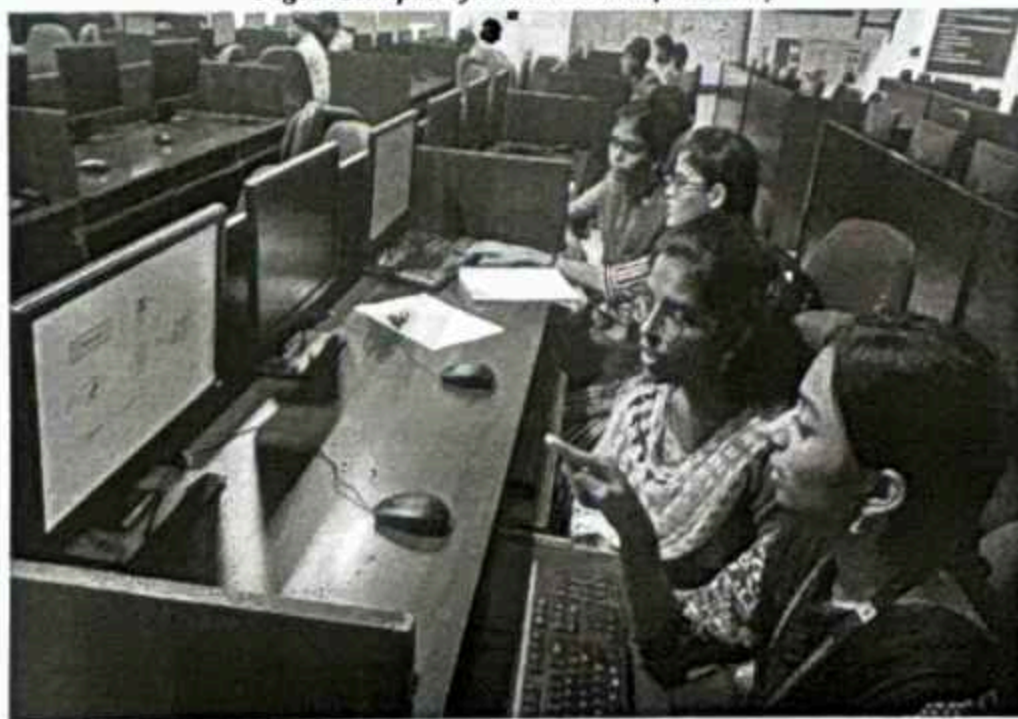


Fig:2 Glimpse of Picture Prompt activity

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Vaishnavi J shared her thought that the visual aspect of the picture prompt captured their attention and stimulated their curiosity about Bitcoin mining.
- Students reported that the activity helped them see the real-world application of blockchain and Bitcoin mining technologies, making the subject matter more relevant and tangible.

Benefit of the practice:

- The activity promoted active learning, encouraging students to actively participate, think critically, and engage with the material rather than passively receiving information.
- The group activity fostered teamwork and collaboration and it allowed students to learn from each other and appreciate different viewpoints.

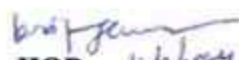
Challenges faced in implementation:

- Some students initially struggled with the lack of background information, which required them to draw heavily on their prior knowledge and understanding.
- Some students grasped the concepts quickly, others needed additional explanations and support to fully understand the image and its relevance to Bitcoin mining.

References:

1. https://biology.mit.edu/wpcontent/uploads/2020/12/TA_Curated_Active_Learning_Techniques.pdf
2. <https://li.wsu.edu/documents/2018/01/student-engagement-strategies-picture-prompt.pdf/>


Signature of Faculty Member


HOD